The Determinants of Capital Structure

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> *Assignment.* Download annual information on North American companies from WRDS for 2010–2020, including the debt ratio and other variables that determine corporate performance. (1) Clean and describe the database, describe the variables, present summary statistics for each variable, and illustrate the distribution of the outcome. (2) Show the correlation between the outcome and the predictors. (3) Write down and explain a linear model of the debt ratio, estimate it by OLS, identify the important factors, interpret the coefficients and report the R-squared.

Summary

- **Data.** 36,718 firm-years for 5,822 US and Canadian non-financial firms, FY2010–2020.
- **Determinants.** Larger firms, firms with more tangible assets and firms in high-leverage industries carry more debt. Profitable, cash-rich and dividend-paying firms carry less.
- **Fit.** OLS explains **21.4%** of the variation in the debt ratio, and **24.8%** with industry and year effects.

Question 1. The database: cleaning, variables and summary statistics

1.1 What kind of database is it?

- **Source. Compustat North America Fundamentals Annual** (`comp.funda`), accessed through Wharton Research Data Services. The data are pulled in code using the WRDS Python API, with Compustat's standard screens: industrial format, standardized data, domestic population and consolidated statements.
- **Structure.** The data form an **unbalanced panel**: each row is one firm (identified by `gvkey`) in one fiscal year. Firms enter and leave as they list, merge or delist, so the average firm appears in 6.3 of the 11 years.

1.2 How was it cleaned, and how many observations remain?

Step	Firm-years left	Why
Compustat records, FY2009–2020	136,665	FY2009 is kept only to build lagged predictors
Incorporated in the US or Canada	123,808	Assignment asks for North American companies
Drop financials (SIC 6000–6999) and utilities (SIC 4900–4949)	70,523	Debt is part of a bank's business model, and utility leverage is set by regulators
Total assets ≥ $10M, positive sales, debt items reported	48,070	Micro-firms have a mean debt ratio of 9.8, a shell-company artifact
Prior-year data available (one-year lag)	40,283	Every predictor is measured at *t − 1*
FY2010–2020, complete cases	**36,718**	**Final estimation sample, 5,822 firms**

Further cleaning steps:

- **No duplicate firm-years** exist, so the panel index (`gvkey`, `fyear`) is unique.
- **R&D** is missing for 44% of firm-years, mostly outside R&D-intensive industries, so a blank means "no material R&D". It is set to 0, with a `rd_missing` flag.
- **Market-to-book** cannot be computed without a share price (9% of rows). These rows are dropped rather than imputed.
- **Outliers.** All ratios, *including the debt ratio*, are winsorized at the 1st and 99th percentiles. Before winsorizing, market-to-book reached 29,351 and profitability −6.3.
- **Sanity check.** Apple's FY2020 values match its 10-K: debt ratio 0.38, EBITDA/assets 0.23 and cash/assets 0.28.

1.3 The variables

Variable	Compustat definition	Expected sign	Economic rationale
Debt ratio (outcome)	(DLTT + DLC) / AT		Share of assets financed with debt
Size	log(AT)	+	Larger firms are diversified and face lower distress costs (trade-off theory)
Tangibility	PPENT / AT	+	Fixed assets can be pledged as collateral
Profitability	OIBDP / AT	–	Profitable firms use retained earnings first (pecking-order theory)
Market-to-book	(AT – CEQ + CSHO × PRCC_F) / AT	–	Growth options lose value in distress
Cash	CHE / AT	–	Liquid assets substitute for borrowing
R&D intensity	XRD / AT	–	Intangible investment supports little debt
Dividend payer	1 if DVC > 0	–	Mature firms with good access to internal funds
Industry leverage	Median debt ratio of the 2-digit SIC industry-year	+	Industry-wide debt capacity and norms

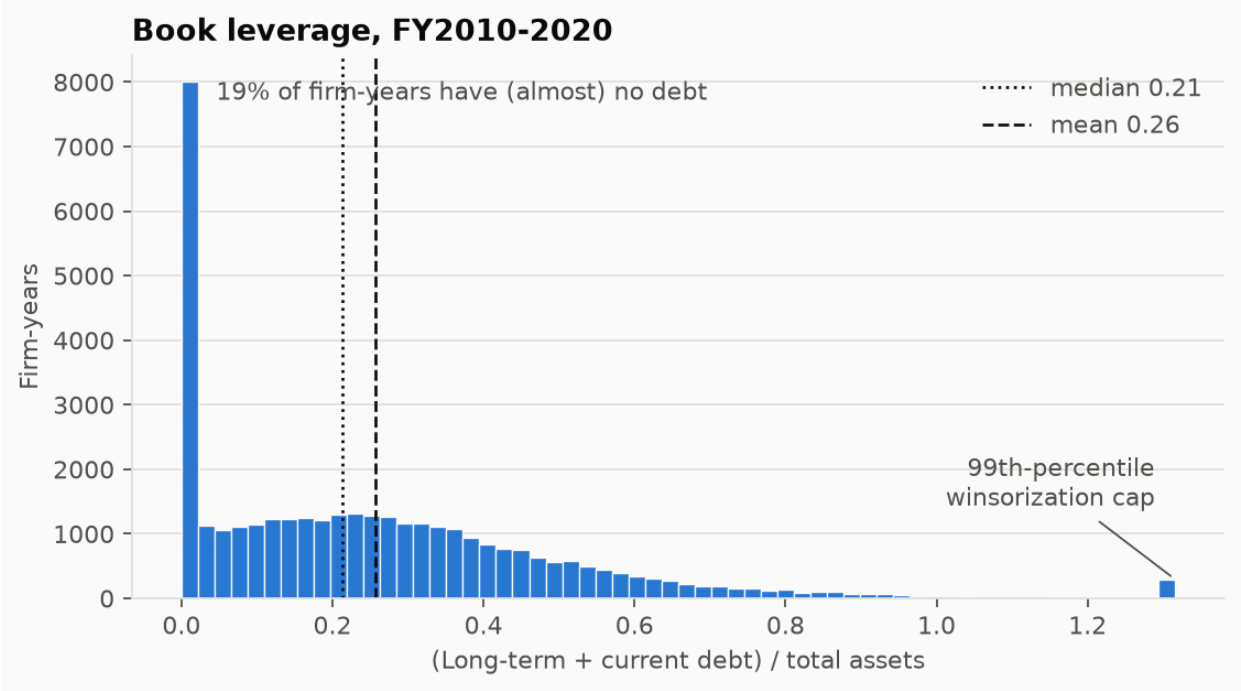
Controls are the `rd_missing` flag and a flag for firms reporting in Canadian dollars.

1.4 Summary statistics

Variable	N	Mean	SD	Min	25%	Median	75%	Max
Debt ratio	36,718	0.257	0.247	0.000	0.044	0.214	0.384	1.315
Size (log assets)	36,718	6.255	2.075	2.303	4.661	6.204	7.702	13.614
Tangibility	36,718	0.285	0.268	0.001	0.069	0.183	0.444	0.944
Profitability	36,718	0.052	0.203	–0.996	0.023	0.097	0.151	0.403
Market-to-book	36,718	2.012	1.560	0.471	1.107	1.500	2.289	10.107
Cash	36,718	0.195	0.222	0.000	0.035	0.109	0.272	0.939
R&D intensity	36,718	0.049	0.107	0.000	0.000	0.000	0.045	0.680
Dividend payer	36,718	0.356	0.479	0	0	0	1	1
Industry leverage	36,718	0.206	0.109	0.000	0.121	0.196	0.273	0.945

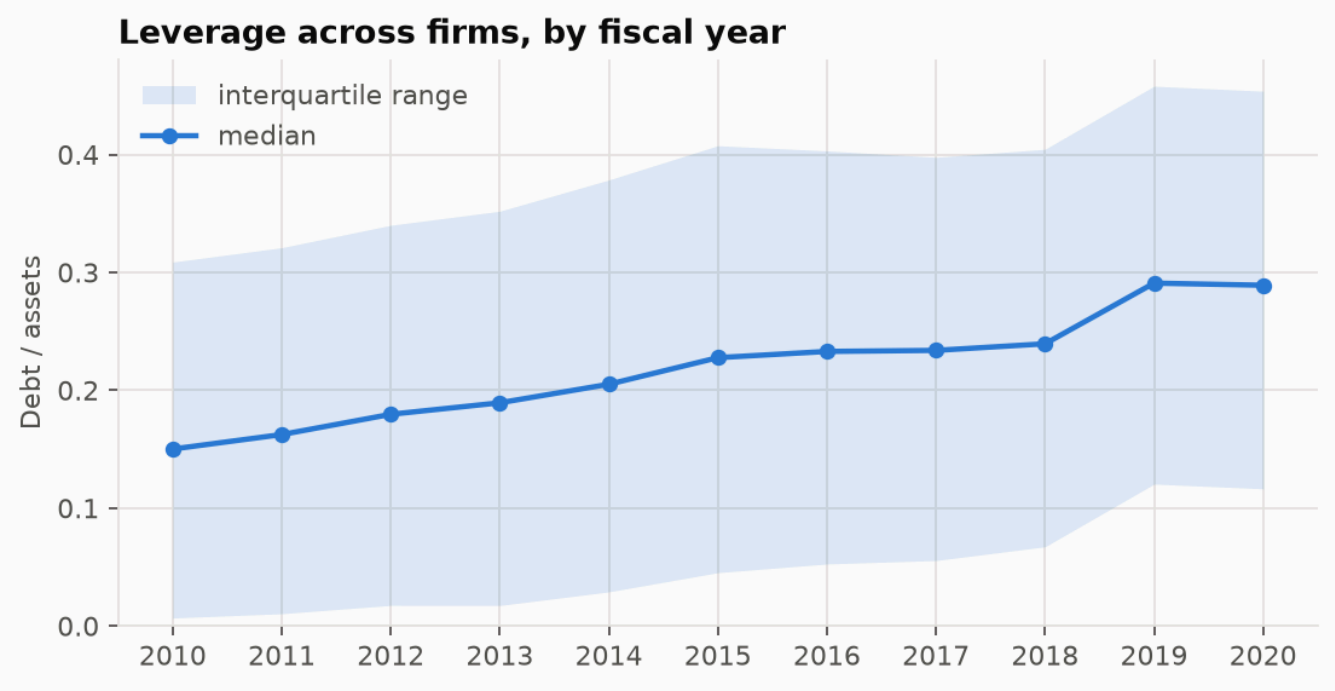
- **Typical firm.** The median firm has about $500M of assets (e^6.2), finances 21% of them with debt, and holds 11% in cash.
- **Profitability.** Median EBITDA is 10% of assets. The mean is only 5% because a tail of loss-makers pulls it down.
- **Dividends.** About a third of firms pay dividends.

1.5 Distribution of the outcome



The debt ratio is right-skewed, with a median of 0.21 and a mean of 0.26.

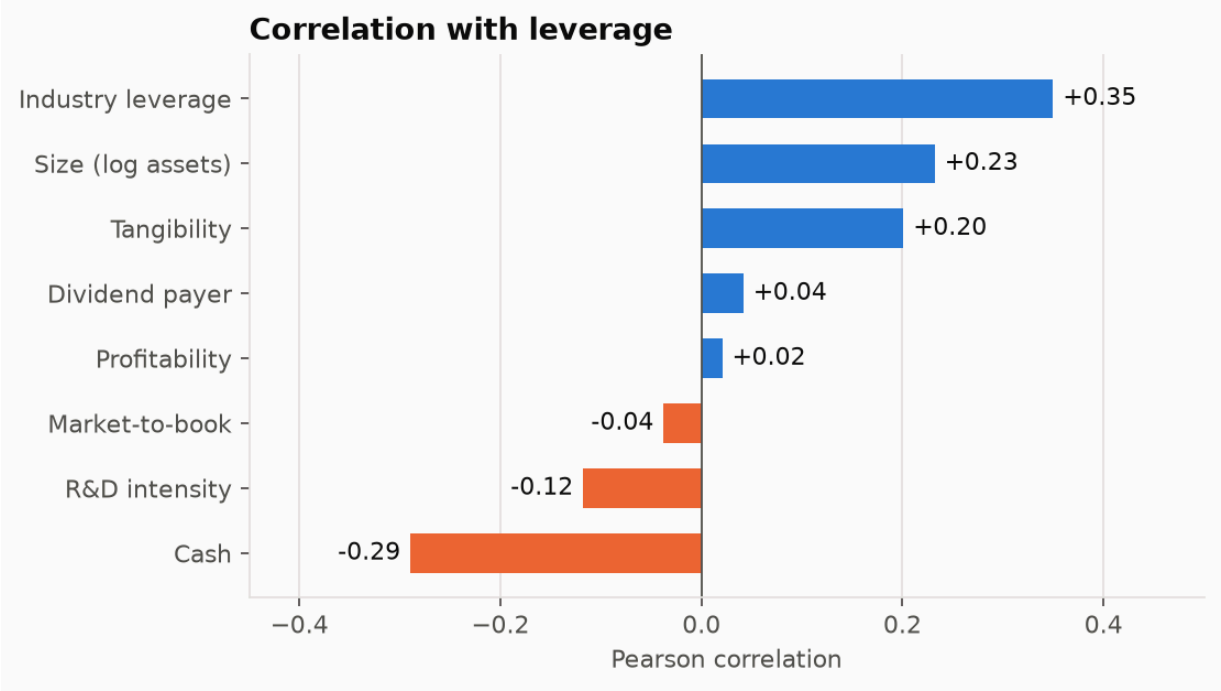
- **No debt.** 19% of firm-years carry essentially no debt, a well-known group of young, cash-rich and R&D-intensive firms.
- **Negative equity.** A small mass sits at the winsorization cap of 1.31: firms whose debt exceeds their book assets.



The median debt ratio rose from 0.15 in 2010 to 0.29 in 2019–2020. The step in 2019 coincides with new lease-accounting rules that brought leases onto balance sheets. Shifts like this affect all firms at once, which is why the model includes year effects.

Question 2. Correlation between the debt ratio and the predictors

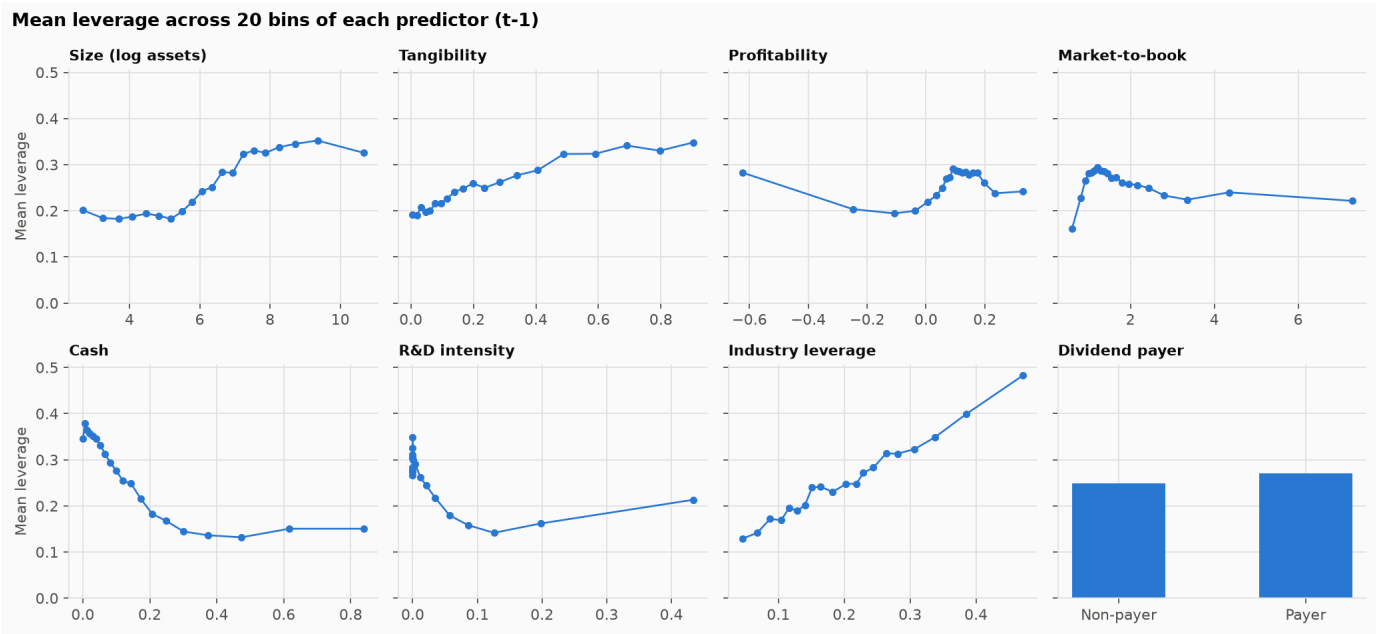
	Industry leverage	Size	Tangibility	Dividend payer	Profitability	Market-to-book	R&D intensity	Cash
Correlation with debt ratio	+0.35	+0.23	+0.20	+0.04	+0.02	−0.04	−0.12	−0.29



Correlations with the debt ratio.

- **Positive:** industry leverage, size and tangibility. Firms borrow more when their industry does, when they are large, and when they have collateral.
- **Negative:** cash and R&D intensity.
- **Near zero:** profitability. The relationship is U-shaped (below), and profitable firms tend to be large (correlation 0.38), and size raises leverage. Only the regression can separate the two.

Correlations among the predictors. The largest is −0.62, between profitability and R&D intensity. All variance inflation factors are below 2.3, far under the usual threshold of 10, so multicollinearity is not a concern.



Each panel shows mean leverage in 20 equal-sized bins of a predictor.

- **Close to linear:** tangibility and industry leverage.
- **Strongly decreasing:** cash, until it reaches about 30% of assets.
- **Threshold:** size matters only above roughly $250M of assets.
- **Non-monotonic:** profitability.

Question 3. A linear model of the debt ratio

3.1 Specification

$$\begin{aligned}\text{DebtRatio}_{it} = \; & \alpha + \beta_1 \,\text{Size}_{i,t-1} + \beta_2 \,\text{Tangibility}_{i,t-1} + \beta_3 \,\text{Profitability}_{i,t-1} \\ & + \beta_4 \,\text{MarketToBook}_{i,t-1} + \beta_5 \,\text{Cash}_{i,t-1} + \beta_6 \,\text{R\&D}_{i,t-1} \\ & + \beta_7 \,\text{DividendPayer}_{i,t-1} + \beta_8 \,\text{IndustryLeverage}_{i,t-1} \\ & + \gamma' \,\text{Controls}_{i,t-1} + \mu_j + \lambda_t + \varepsilon_{it}\end{aligned}$$

- ***i* and *t*** index firms and fiscal years.
- **Each β** is the change in the debt ratio from a one-unit change in that characteristic, holding the others fixed.
- **Lagged predictors.** They are measured at *t − 1*, so they are determined *before* the debt ratio they explain. This limits reverse causality, for example a debt issue that raises cash in the same year.
- μ_j **and** λ_t are industry (2-digit SIC) and year effects. They absorb industry norms and economy-wide shocks.

The model is estimated by **OLS** in three versions:

1. **Pooled OLS**, without μ_j and λ_t.
2. **OLS with industry and year dummies.** This is the preferred specification.
3. **Firm and year fixed effects.** A robustness check that uses only changes within each firm over time.

Because each firm appears in several years, standard errors are **clustered by firm**.

3.2 Estimates

	(1) Pooled OLS	(2) + Industry & year	(3) Firm & year FE
Size	0.0224*** (0.0015)	0.0236*** (0.0014)	0.0211*** (0.0046)
Tangibility	0.0342*** (0.0127)	0.1190*** (0.0178)	0.1166*** (0.0263)
Profitability	−0.1697*** (0.0173)	−0.1775*** (0.0170)	−0.0979*** (0.0181)
Market-to-book	0.0166*** (0.0021)	0.0147*** (0.0020)	−0.0004 (0.0019)
Cash	−0.3134*** (0.0152)	−0.3238*** (0.0156)	−0.1323*** (0.0169)
R&D intensity	0.0999*** (0.0380)	0.0198 (0.0382)	−0.0396 (0.0490)
Dividend payer	−0.0471*** (0.0056)	−0.0490*** (0.0056)	0.0004 (0.0053)
Industry leverage	0.5902*** (0.0289)	0.3911*** (0.0384)	0.2868*** (0.0350)
R^2	**0.214**	**0.248**	**0.034** (within)
Observations	36,718	36,718	36,718

Firm-clustered standard errors are in parentheses. *** $p < 0.01$. All specifications include the R&D-missing and Canadian-dollar controls. In (3), the Canadian-dollar flag is absorbed by the firm effects.

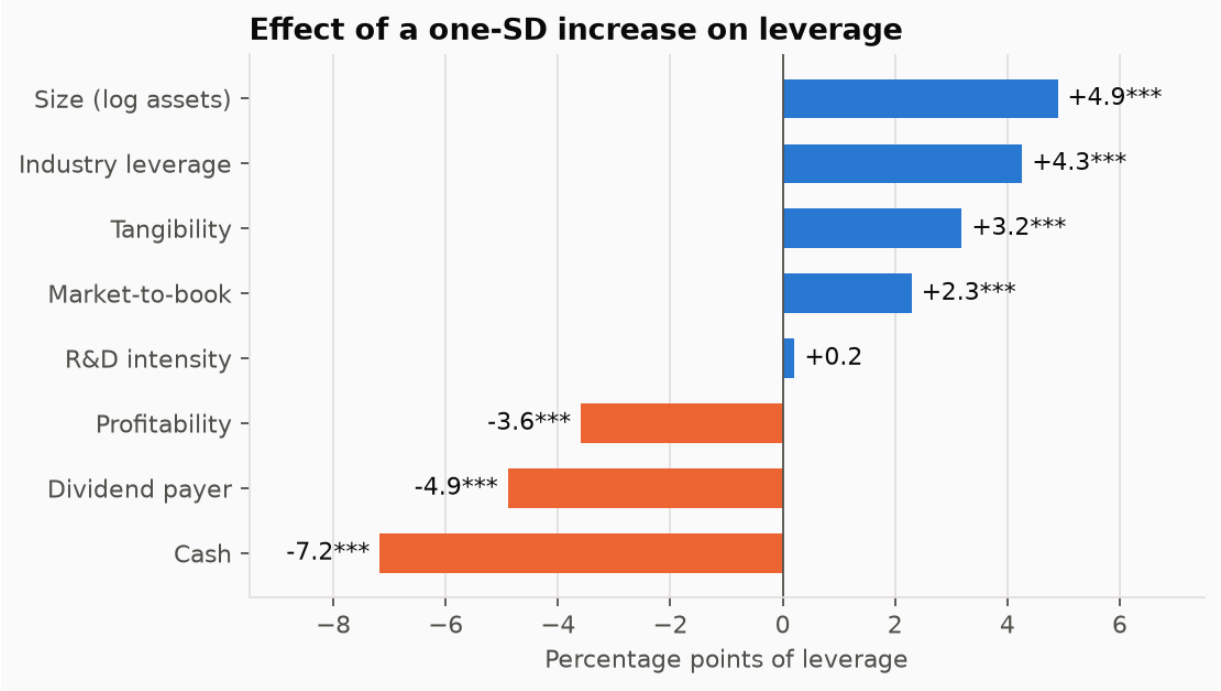
3.3 How to interpret the coefficients

These interpretations use specification (2). The debt ratio is a fraction, so a coefficient of 0.10 means 10 percentage points (pp).

Variable	Coefficient	Reading
Size	+0.024	Doubling total assets raises the debt ratio by **1.6 pp** (0.024 × ln 2)
Tangibility	+0.119	10 pp more of assets in PP&E → debt ratio **+1.2 pp**
Profitability	−0.178	EBITDA/assets 10 pp higher → debt ratio **−1.8 pp**
Market-to-book	+0.015	Market value one times book assets higher → debt ratio **+1.5 pp**
Cash	−0.324	10 pp more of assets in cash → debt ratio **−3.2 pp**
R&D intensity	+0.020	Not significant once industry is controlled for
Dividend payer	−0.049	Payers have a debt ratio **4.9 pp lower** than comparable non-payers
Industry leverage	+0.391	Industry median 10 pp higher → firm's debt ratio **+3.9 pp**

3.4 Which factors matter most?

The units differ, so the chart compares the effect of a **one-standard-deviation** increase in each predictor.



1. **Cash (−7.2 pp per SD)** has the largest effect. Firms with liquid assets need less external finance.
2. **Size (+4.9 pp)** supports the trade-off theory: large firms have more stable cash flows and lower expected bankruptcy costs.
3. **Dividend payers (−4.9 pp)** are mature firms that fund themselves internally. In the raw data payers look slightly *more* levered, because they are large. The regression removes this confounding.
4. **Industry leverage (+4.3 pp)** and **tangibility (+3.2 pp)**. Industry norms and collateral both matter.
5. **Profitability (−3.6 pp)** supports the **pecking-order theory**: profitable firms use retained earnings instead of borrowing. It contradicts the simple trade-off prediction.

Two results do not match theory:

- **Market-to-book (+2.3 pp)** has the opposite sign in pooled data and disappears within firms (3). A likely reason is that firms with depressed or negative book equity mechanically have both high debt ratios and high market-to-book.
- **R&D intensity** loses significance once industries are compared like for like.

Robustness (3). Size, tangibility, profitability, cash and industry leverage keep their signs and significance with firm fixed effects. The firm effects are jointly significant ($F = 14.6$, $p < 0.001$).

3.5 What is the R-squared?

- **Pooled OLS:** R^2 is **0.214**, so the characteristics explain 21% of the variation in debt ratios across firm-years. The F-statistic of 236 ($p < 0.001$) rejects the hypothesis that all slopes are zero.
- **With industry and year dummies:** R^2 rises to **0.248**.
- **Firm fixed effects:** the *within* R^2 is 0.034. Year-to-year changes in a firm's characteristics explain little of its year-to-year changes in leverage.

This is a well-documented feature of capital structure: debt ratios are highly persistent, and most of their variation is a stable firm-specific component (Lemmon, Roberts & Zender 2008). The pooled R^2 captures the differences *between* firms that the fixed effects absorb.

3.6 Limitations

- **Association, not causation.** The estimates are conditional associations, not causal effects. Lagging the predictors limits simultaneity but not omitted-variable bias.
- **Bounded outcome.** The debt ratio cannot fall below zero, and 1.8% of fitted values do. A fractional-response model would respect the bound. The linear model fits well across the middle of the distribution.
- **Scope.** Results apply to non-financial firms with at least $10M of assets.

Full code, including every cleaning step: `notebooks/01_capital_structure.ipynb`

References.

- Frank, M. & Goyal, V. (2009). Capital structure decisions: Which factors are reliably important? *Financial Management* 38(1).
- Lemmon, M., Roberts, M. & Zender, J. (2008). Back to the beginning: Persistence and the cross-section of corporate capital structure. *Journal of Finance* 63(4).
- Rajan, R. & Zingales, L. (1995). What do we know about capital structure? Some evidence from international data. *Journal of Finance* 50(5).